

Factorial function and Wilson's thm.

- **Def.** Factorial function is denoted $n!$ for n a non-negative integer the product of all positive integers less than or equal to n
 - *Equivalently* can define inductively:
 - $0! = 1$
 - $(n + 1)! = (n + 1) \cdot n!$
- **Ex:** $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$
- **Wilson's Thm.** For all p prime, $(p - 1)! \cong -1 \pmod{p}$
- **Pf.** Let's consider each number $x \in \{1, \dots, p - 1\}$
 - First note that every number $x \in \{1, \dots, p - 1\}$ is rel. prime to p
 - This means that every number $x \in \{1, \dots, p - 1\}$ has a **unique mult. Inverse** mod p
 - Now we **claim:** in this list, only 1 and $p-1$ are *self-inverse* mod p – why?
 - i.e., $x^2 \cong 1 \pmod{p} \Rightarrow x^2 - 1 \cong 0 \pmod{p} \Rightarrow p \mid (x - 1)(x + 1)$
 - So this means $x \cong +1 \pmod{p}$ or $x \cong -1 \pmod{p}$ – which numbers satisfy?
 - Out of the list, only satisfied by $x = 1$ or $x = p - 1$
 - All the other elements in the list have a non-self mult. Inverse
 - Now consider $(p - 1)! = (p - 1) \cdot (p - 2) \cdot \dots \cdot 1$
 - The “middle” terms from $\{2, \dots, p - 2\}$ can be reordered into distinct pairs of numbers and multiplicative inverses.
 - These pairs each multiply to 1, and disappear, leaving us with 1 and $p - 1$. And $1 \cdot p - 1 = p - 1 \cong -1 \pmod{p}$
QED.

Another very helpful lemma

- **Helpful lemma 2:** Let p be prime, and a be such that $\gcd(a, p) = 1$. Then, as sets, $\{a \cdot 1, a \cdot 2, \dots, a \cdot (p - 1)\} = \{1, 2, \dots, p - 1\}$ (where multiplication here is done mod p)
- **Pf.**
 - As $\gcd(a, p) = 1$ there must exist a mult. Inverse a^{-1} for a .
 - Then we can recover the set on RHS by multiplying each element in the set on LHS by a^{-1} -- i.e., there's a one-to-one correspondence between these two sets. QED

Fermat's Little Theorem

- **Thm.** For all primes p , and a such that $\gcd(a, p) = 1$, $a^{p-1} \cong 1 \pmod{p}$
- **Pf.** Consider the product $n = (a \cdot 1) \cdot (a \cdot 2) \cdot \dots \cdot (a \cdot (p-1)) \pmod{p}$
 - On the one hand, by **helpful lemma 2**, this is just a reordering of the factorial, and so $n = -1$, by Wilson's Thm. (i.e., this is $= (p-1)! = -1 \pmod{p}$).
 - On the other hand, by factoring out the a 's (of which there are $p-1$), we have $n \cong a^{p-1} \cdot (p-1)! \pmod{p}$
 - So we have $n \cong a^{p-1} \cdot (-1) \pmod{p}$
 - by Wilson's thm. (used here again!)
 - Equating with $n = -1$, and multiplying both sides by -1 gives the Thm. QED

Who cares about Fermat's Little Theorem (or number theory, more generally?)

- Historically, number theorists were proud of studying the “purest” form of math
 - In the sense that it didn't really have any applications to the real world.
- Today, we know that this is not true at all.
- Possibly most interesting connection of number theory to CS: Cryptography!